Exact Scholten-to-weighted-model normalization 2026-10-31

T-1130. Producer proof packet; independent source-fidelity review pending.

Checked model identity

The source's fixed Gaussian constants give the quintic

$$F(t) = 1 + \frac{25}{3}t + \frac{299}{12}t^2 + \frac{95}{3}t^3 + \frac{187}{12}t^4 + \frac{5}{2}t^5.$$

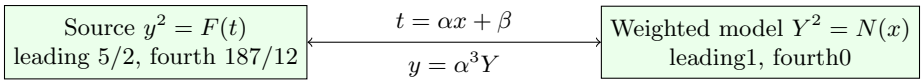
The historical normalization helper uses $\alpha = 5/2$, $\beta = -187/150$ and ordinate scale $\alpha^3 = 125/8$. Lean checks

$$F\left(\frac{5}{2}x - \frac{187}{150}\right) = \left(\frac{125}{8}\right)^2 N(x),$$

where

$$N(x) = x^5 - \frac{12938}{28125}x^3 + \frac{907312}{10546875}x^2 + \frac{44207039}{1318359375}x - \frac{57913058672}{7415771484375}.$$

The field hypotheses are $2, 3, 5 \neq 0$; source expansion also assumes $i^2 = -1$. All denominator nonzeroness is explicit. These hypotheses cover the declared source primes $p > 13$, without a new primality or cohort run.



Formal scope

Twelve main declarations establish source expansion, normalization, both coordinate round trips, curve-equation transport, evaluation in the existing `WeightedDoubling.curve` definition, and coefficients4/5 equal to0/1. Its parameter order is (a_0, a_1, a_2, a_3).

Eight fixed controls check $F(0) = 1$, $F(-1/2) = 0$, coordinate values, wrong ordinate scaling, round trip and the helper's shift/scale. The $(t, y) = (3, 7)$ fixture tests coordinate arithmetic; it is not asserted to lie on the curve. All20 declarations receive separate axiom readback. Fraction-pair polynomial expansion provides an additional producer cross-check, not independent review.

Connection to existing geometry

The weighted-Proj and infinity-stalk modules already establish geometric properties of their model. Their Pic0Bridge explicitly does not construct a Picard group or a global class equality. This packet identifies the source's affine equation with that formula model. Global projective extension, squarefreeness transport, the literal Mumford substitution loops and Picard transport are next bindings; the affine identities do not silently supply them.

Evidence and sources

Attempt1 failed on power rewriting/tactic sequencing; attempt2 passed. Successful inputs remain bound; linter warnings and the failure are retained. Source references: `three_leads.rs`, lines1094–1171; historical `genus2_model_gate.rs`, lines14–56 and75–88. Primary documentation: Mathlib polynomial evaluation and composition. No novelty claim.

No scientific execution, timing or cost ratio; expired campaign limits unchanged. No independent binary/kernel rechecker or independent source-fidelity review is claimed. Full proof sources, raw receipts, coefficient JSON and source/custody hashes accompany this report.